

the global commercial real estate services firm. He adds that the demand for data centres will double globally, driven by AI.

But there's a catch here. These data centres consume massive amounts of electricity.

Data from the International Energy Agency (IEA) bears this out. In 2022, data centres consumed 460 terawatt hours (TWh) of electricity, or 2% of all global electricity usage. That is expected to double now, thanks to the increasing adoption of large language models (LLMs) for AI. The result is a likely increase in emissions from data centres that contribute around 2.5–3.7% of all global emissions, according to carbon intelligence start-up Climatiq.

It's not a surprise, then, that countries are looking at ways to make these data centres green. India—which is still in the early stages of the boom, ranked 14th in the world according to research firm Statista with 163 facilities, compared to 5,381 in the US—is also looking at mitigation measures.

BEARING THE LOAD

India's data centre capacity is expected to exceed 1,300 MW by the end of 2024 from 880 MW as of June 2023, according to real estate consulting firm CBRE South Asia. In line with this, the industry's power consumption is expected to account for 1.5–2% of the nation's total power consumption.

And though the country is scaling up renewable energy capacity, most of the increase in demand is likely to be met through fossil fuel, which still accounts for 57% of India's installed generation capacity as of February, per the Ministry of Power.

One option that companies have is to optimise the type and quantity of power consumed. That's where the Green Open Access policy introduced by the government in 2022 comes in. It allows entities consuming over 1 MW, like data centres, to directly source green energy from private producers.

"Under these agreements, producers supply a specified amount of green energy to the National Grid on behalf of the data centre. When the data centre draws power from the grid, its utility provider bills only for the difference between the consumed and supplied green energy units," says Sunil Gupta, Co-Founder and CEO of Yotta Data Services, a Hiranandani Group firm that builds data centres. "This allows any data centre, whether it's a green-field set-up or existing facility, to operate with a significantly reduced carbon footprint." For instance, Yotta's Navi Mumbai data centre obtains over 50% power from hydroelectric sources through this.

TOWARDS GREEN POWER

- 1 **Estimated electricity consumption by data centres globally in 2022 was around 2% of total demand; the figure is set to double by 2026**
- 2 **Data centre emissions, including cloud computing, account for around 2.5–3.7% of all global emissions**
- 3 **India's data centre capacity is expected to exceed 1,300 MW by the end of 2024, from 880 MW as of June 2023**
- 4 **The Indian data centre industry's power consumption is expected to account for 1.5–2% of the nation's total power consumption**
- 5 **Operational and cooling needs are the biggest contributors to data centres' carbon footprint**
- 6 **The cost of building a green data centre varies based on factors like size, location, and tech, and could result in a 5-15% premium upfront**

SOURCE IEA, CLIMATIQ, CBRE, BT RESEARCH

GLOBAL FOOTPRINT

● The US dominates the data centre arena with 5,381 facilities ● India ranks 14th

